Core practical – Investigating reaction rates, Part 1

Pearson Edexcel GCSE (9-1) Combined Science / Chemistry

Name _____ **Class** _____ **Date** _____

Part 1: Measuring the production of a gas

Practical Objective

To investigate the effect on the rate of reaction of changing the surface area of solids and the concentration of solutions, by measuring the production of a gas.

Apparatus

- eye protection
- balance
- water trough or beehive shelf
- 100 cm^3 measuring cylinder
- stop clock
- conical flask
- delivery tube and bung
- marble chips (small)
- marble chips (large)
- dilute hydrochloric acid (1 mol dm^{-3} or 36.5 g dm^{-3})

Safety

- Wear eye protection at all times.
- Care is needed with acid solutions. Wash off splashes immediately.

Your teacher may watch to see if you can:

- carefully control variables during investigations.
- measure change accurately.

Method

Task 1

A Set up the apparatus as shown in the diagram.

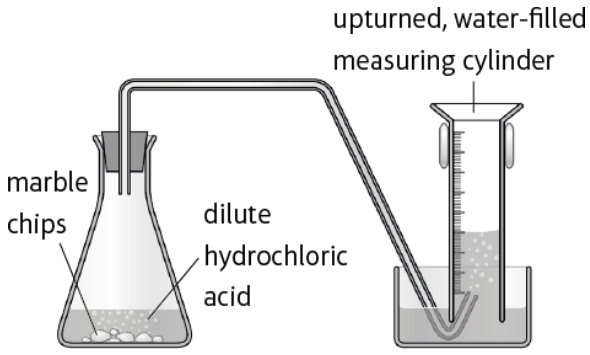
B Measure 40 cm^3 of 1 mol dm^{-3} (36.5 g dm^{-3}) hydrochloric acid into a conical flask.

C Add 5 g of small marble chips to the flask.

D Immediately stopper the flask and start the stop clock/watch.

E Note the total volume of gas produced after every 30 seconds for five minutes or until the reaction has finished.

F Repeat steps **A** to **E** using 5 g of larger marble chips.



Task 2

G Follow steps **A** to **D** above.

H Note the amount of carbon dioxide produced in one minute.

I Repeat steps **G** and **H** using 0.8 mol dm^{-3} (29.2 g dm^{-3}), 0.6 mol dm^{-3} (21.9 g dm^{-3}), 0.4 mol dm^{-3} (14.6 g dm^{-3}) and 0.2 mol dm^{-3} (7.3 g dm^{-3}) acid.

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Recording your results

Task 1

Record your results in the table below.

Time (min)	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
Small chips – volume of gas (cm^3)											
Large chips – volume of gas (cm^3)											

Task 2

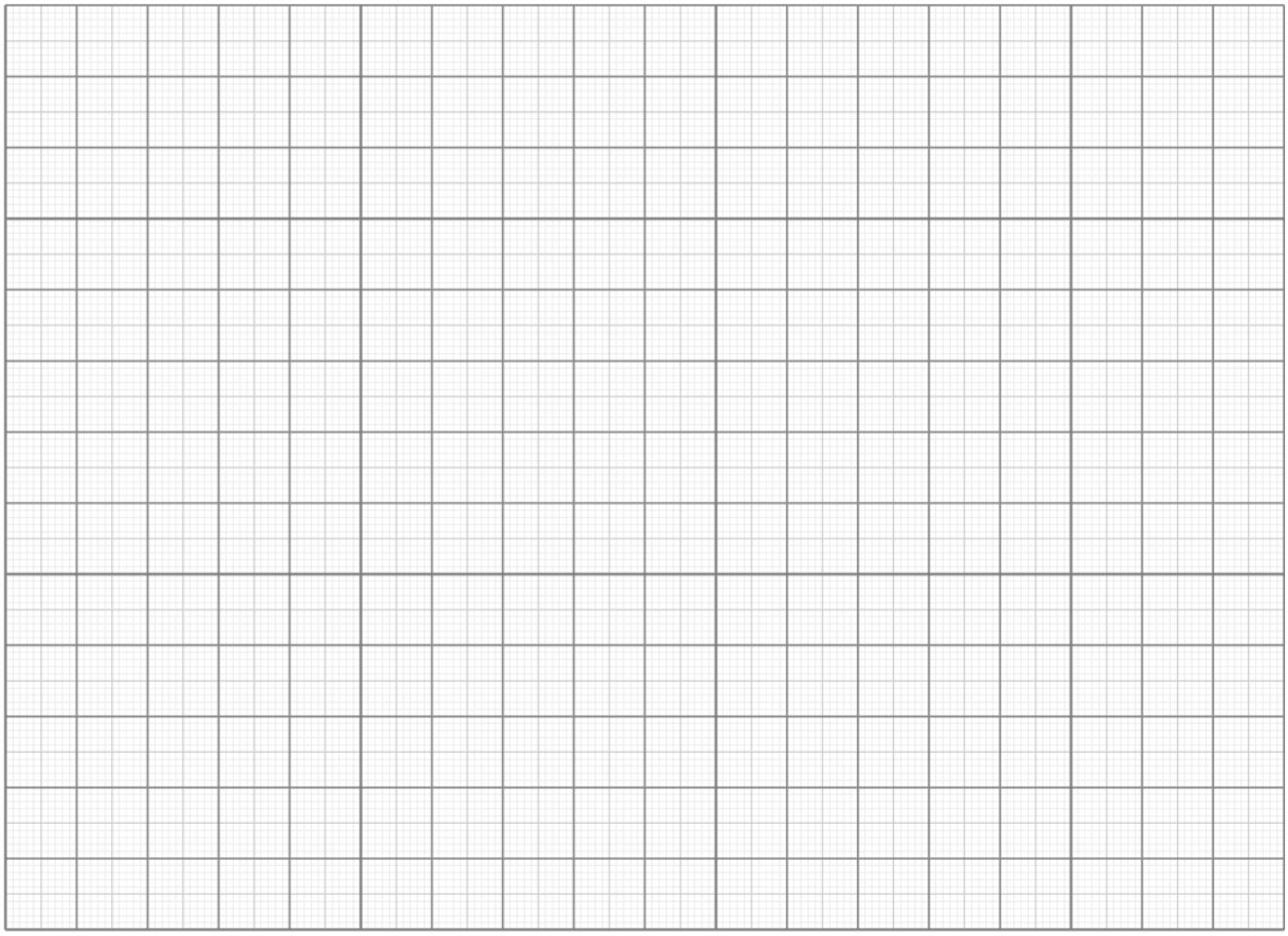
Record your results in the table below.

Concentration of acid ($mol\ dm^{-3}$)	1.0	0.8	0.6	0.4	0.2
Volume of gas produced in one minute (cm^3)					

Considering your results/conclusion

Task 1

1 Use your results to draw a graph of volume of gas against time for small chips and a graph for large chips on the same axes. Put time on the horizontal axis and volume of gas on the vertical axis. Draw a line of best fit for each graph using different coloured lines and labels.



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2 Explain how you can tell from the graphs when the reactions were finished.

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3 Describe how the size of the marble chips is related to the surface area for a fixed mass of chips.

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4 Describe how increasing the surface area affects the rate of reaction.

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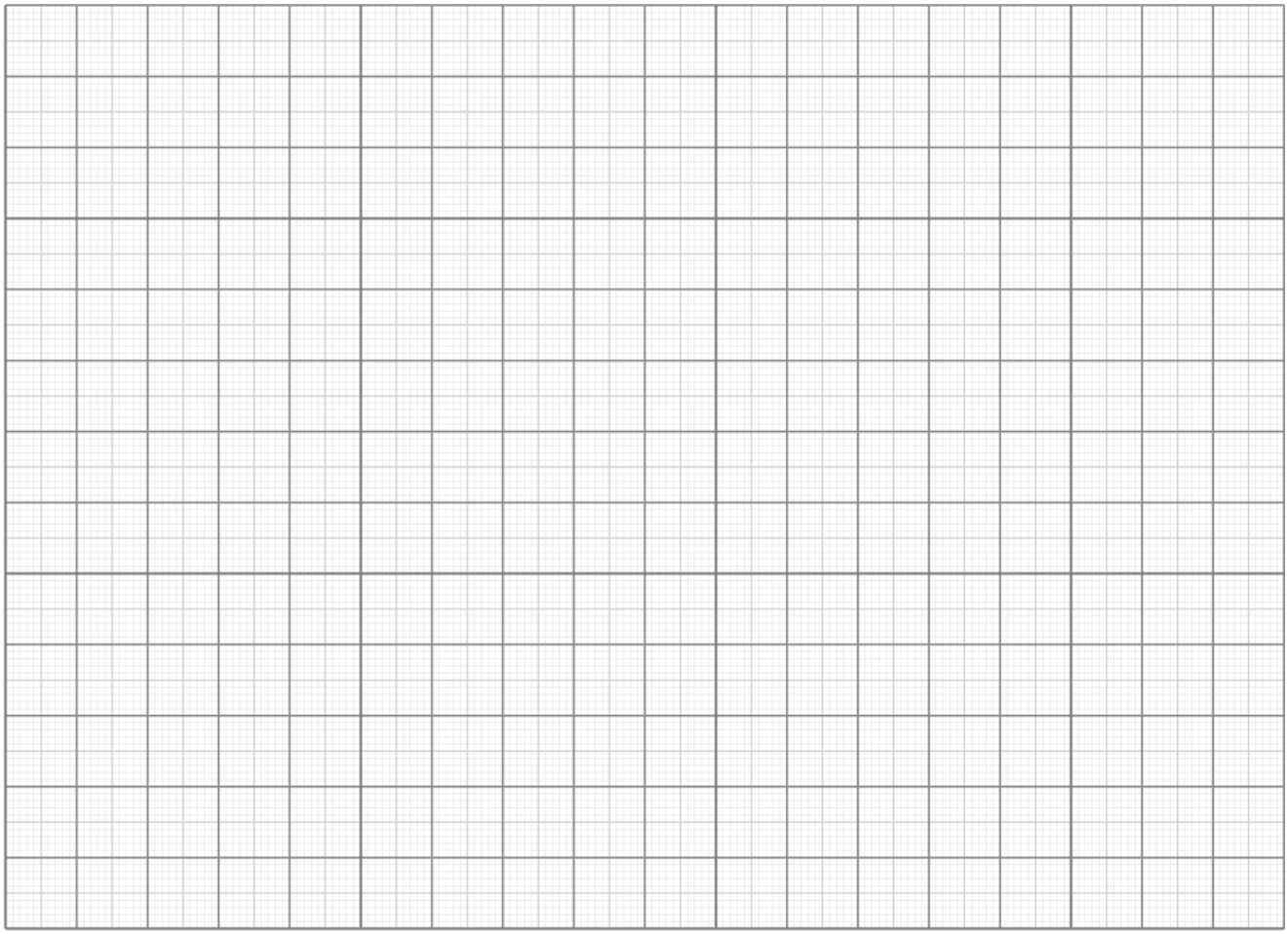
5 Explain how your results and graphs fit with your conclusion in question **4**.

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Task 2

6 Use your results to draw a graph of concentration of acid against volume of gas produced in one minute. Put volume of gas on the horizontal axis and concentration on the vertical axis.



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7 Describe how the rate of the reaction is related to the concentration of the acid.

8 Explain how your results and diagram fit with your conclusion in question **7**.

Evaluation

Tasks 1 & 2

9 Suggest possible sources of error in these investigations.

10 Suggest possible changes to the methods of both tasks that could improve the reliability of the results.

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Core practical – Investigating reaction rates, Part 2

Pearson Edexcel GCSE (9-1) Combined Science / Chemistry

Name _____ **Class** _____ **Date** _____

Part 2: Observing a colour change

Practical Objective

To investigate the effect of changing the temperature on the rate of reaction between sodium thiosulfate and hydrochloric acid, by observing a colour change in the solutions.

Apparatus

- eye protection
- 250 cm^3 conical flask
- 10 cm^3 measuring cylinder
- 50 cm^3 measuring cylinder
- stop clock
- test tube
- test tube rack
- water bath
- white paper with cross
- sodium thiosulfate solution
- dilute hydrochloric acid
- cling film

Safety

- Wear eye protection at all times.
- Care is needed with acid solutions. Wash off splashes immediately.
- Do not exceed 50 °C

Your teacher may watch to see if you can:

- carefully control variables during investigations.
- measure change accurately.
- work safely.

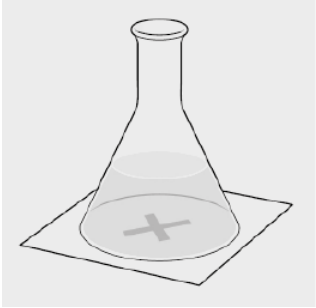
Method

A Decide on four temperatures between 20 °C and 50 °C, which you are going to investigate.

B Place 10 cm^3 of sodium thiosulfate solution and 40 cm^3 of water into a 250 cm^3 conical flask.

C Measure 5 cm^3 of dilute hydrochloric acid into a test tube.

D Clamp the conical flask in place in a water bath at your first chosen temperature. Place the test tube in a rack in the same water bath.



E Record your chosen temperature.

F After five minutes, remove the flask and place it on a piece of white paper marked with a cross, as shown opposite.

G Add the acid to the thiosulfate solution, quickly cover the top of the flask with cling film and start the stop clock.

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H Looking down from above, stop the clock when the cross disappears.

I Note this time and the final temperature of the mixture.

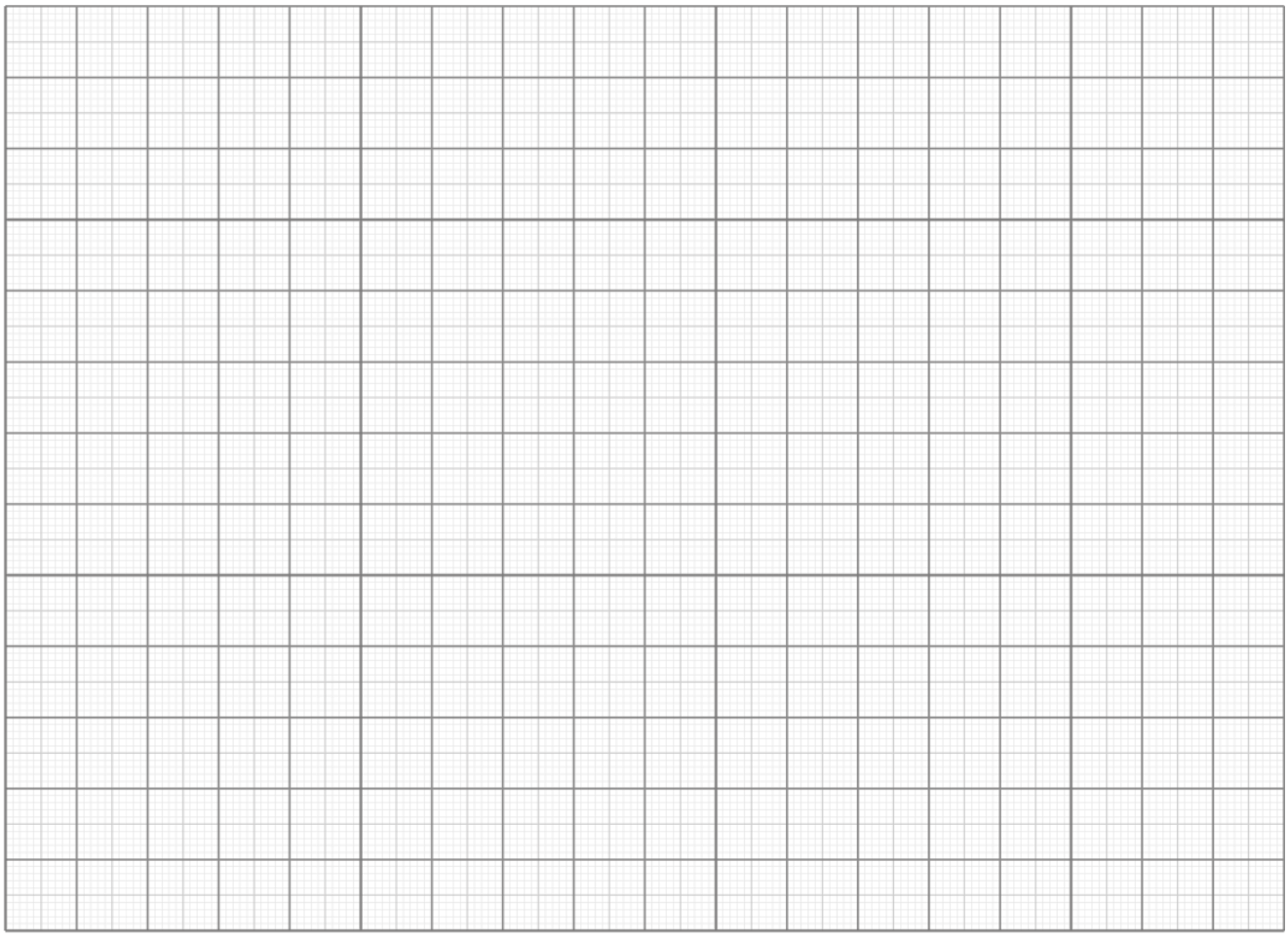
J Repeat steps **A** to **I** for the other chosen temperatures.

Recording your results

1 Draw a table in the space below with two columns: one for temperature, and the other for the time taken for the cross to disappear. Don't forget the units. Record the results of your experiments.

Considering your results

2 Draw a graph of your results, with temperature on the horizontal axis. Draw a line of best fit.



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3 **a** Describe how temperature affects the rate of the reaction.

b Explain your answer to part **a** by referring to the shape of your diagram.

4 If the rate of reaction doubled, what would happen to the time taken for the cross to disappear?

5 **a** What temperature rise roughly doubles the rate of the reaction?

b Use the values from your graph to explain your answer to part **a**.

Evaluation

6 Describe two possible sources of error in this investigation.

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7 Suggest a way of reducing one of these errors.

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